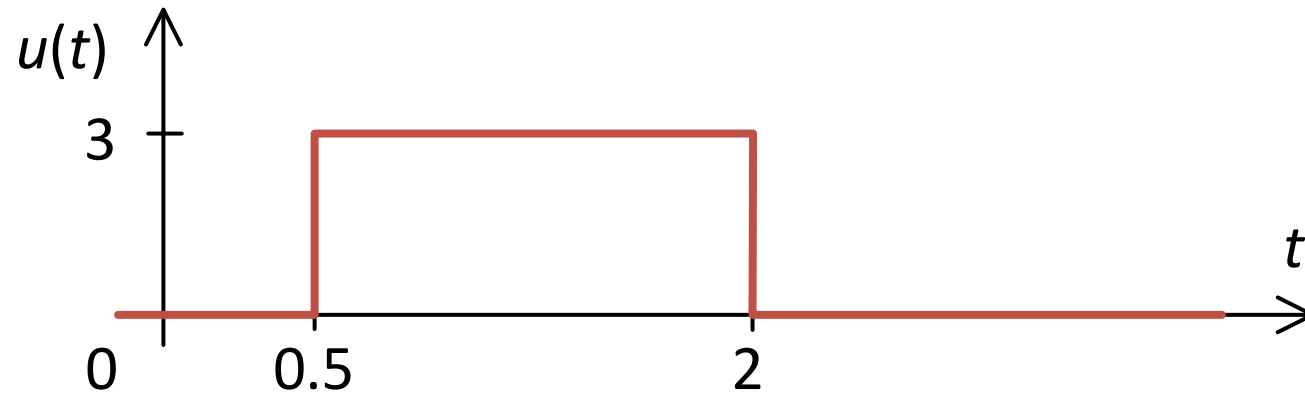


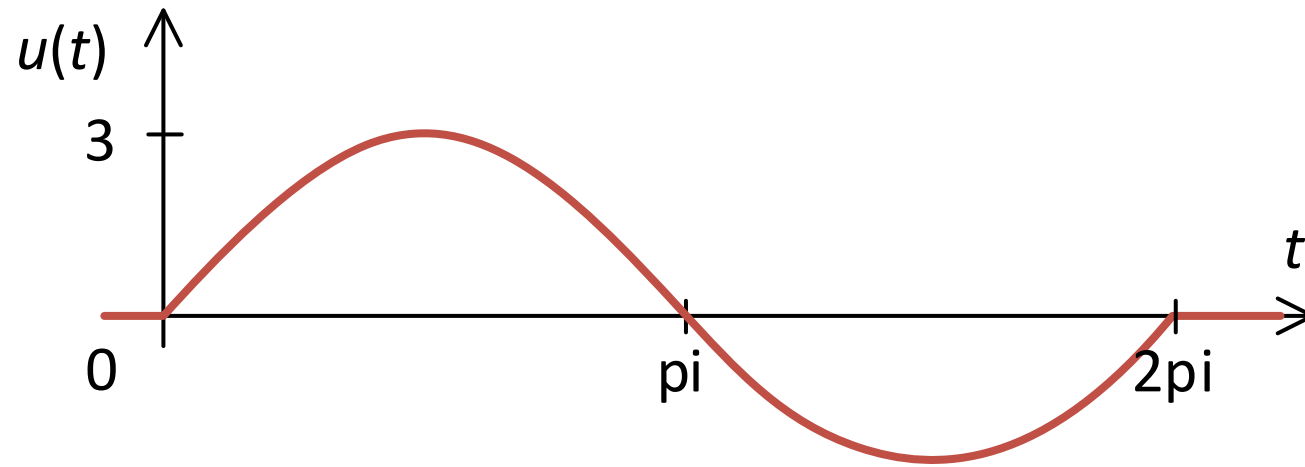
Primena Laplasove transformacije

Neki signali

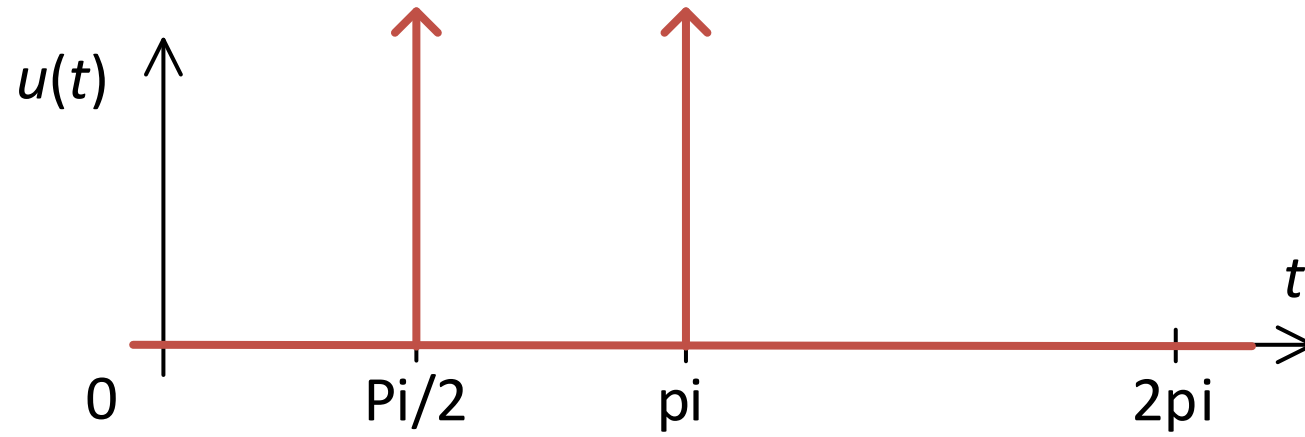
Primer 1



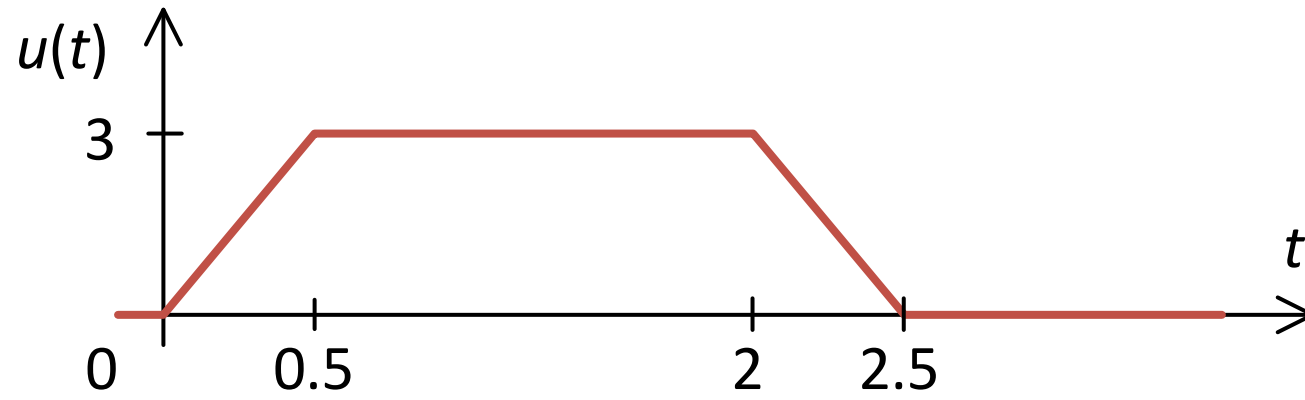
Primer 2



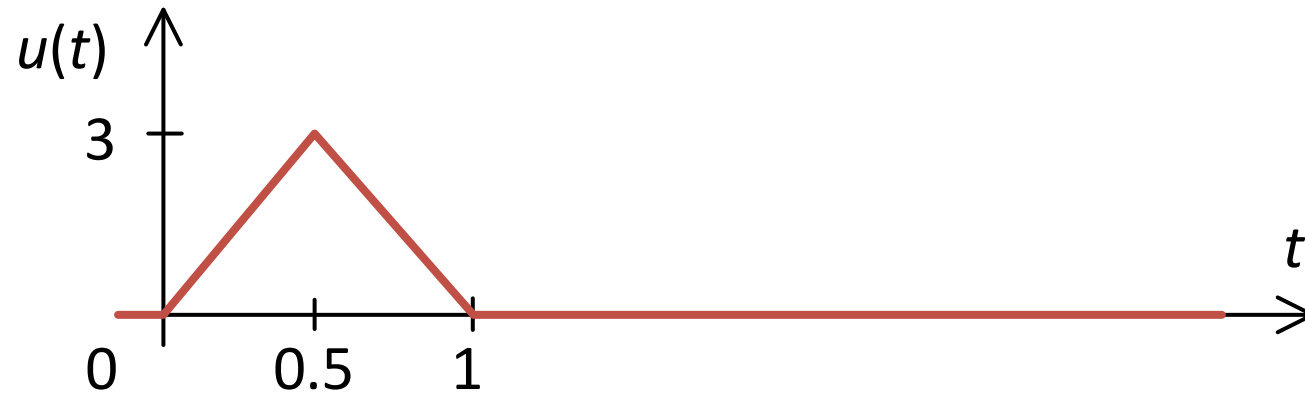
Primer 3



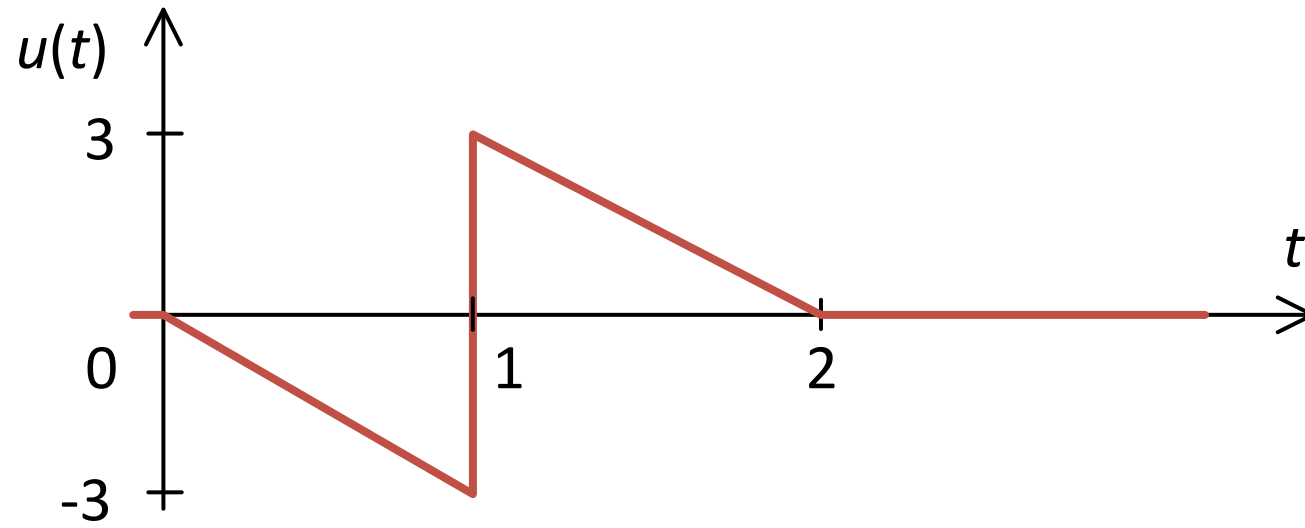
Primer 4



Primer 5



Primer 6



Rešenja

Primer 1: $u(t) = 3h(t - 0.5) - 3h(t - 2)$

$$U(s) = \frac{3}{s} (e^{-0.5s} - e^{-2s})$$

Primer 2: $u(t) = 3 \sin t h(t) - 3 \sin(t - 2\pi) h(t - 2\pi)$

$$U(s) = \frac{3}{s^2 + 1} (1 - e^{-2\pi s})$$

Primer 3: $u(t) = \delta\left(t - \frac{\pi}{2}\right) + \delta(t - \pi)$

$$U(s) = e^{-\frac{\pi}{2}s} + e^{-\pi s}$$

Primer 4: $u(t) = 6th(t) - 6(t - 0.5)h(t - 0.5) - 6(t - 2)h(t - 2) + 6(t - 2.5)h(t - 2.5)$

$$U(s) = \frac{6}{s^2} (1 - e^{-0.5s} - e^{-2s} + e^{-2.5s})$$

Primer 5: $u(t) = 6th(t) - 12(t - 0.5)h(t - 0.5) + 6(t - 1)h(t - 1)$

$$U(s) = \frac{6}{s^2} (1 - 2e^{-0.5s} + e^{-s})$$

Primer 6: $u(t) = -3th(t) + 6h(t - 1) + 3(t - 2)h(t - 2)$

$$U(s) = \frac{3}{s^2} (-1 + e^{-2s}) + \frac{6}{s} e^{-s}$$